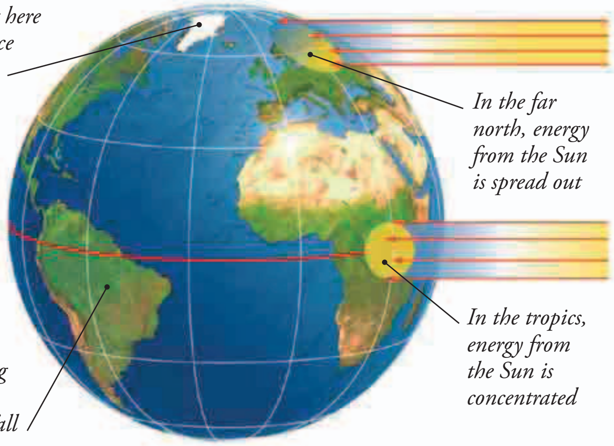
Climate zones

As air currents carry heat away from the tropics, they create warm, wet climate zones near the equator and hot, dry deserts farther north and south. Farther away from the tropics lie the cooler midlatitudes, which are often affected by powerful prevailing winds. The polar regions are cold and dry.



SOLAR ENERGY

Weather is powered by the Sun's energy. This energy is concentrated in the tropics but spreads out over a larger area near the poles. As a result, sunlight becomes less intense as you move away from the equator, which is why Scandinavia is so much cooler than Kenya. The temperature difference drives the global air currents that control rainfall and wind direction.

FAST FACTS

- In the Amazon Rainforest, it rains about 250 days a year.
- It is said that parts of the Atacama Desert in South America have had no rain at all in living memory.
- The average temperature at the North Pole ranges from -17°F (-27°C) in winter to 73°F (23°C) in summer.
- Winds in Antarctica can reach 199 mph (327 km/h).

TROPICAL RAIN

Rising warm, moist air near the equator creates the biggest storm clouds in the world. They may be 10 miles (16 km) high, and are loaded with moisture that falls as heavy tropical rain. The combination of regular rain and year-round warmth is ideal for trees, which form dense rainforests.

SUBTROPICAL DROUGHT

Having lost all its moisture over the tropical forests, high-level air flows north and south, cools, and sinks over the subtropics—the areas just north and south of the tropics. The sinking dry air stops clouds from forming, so there is little rain and lots of sunshine. This creates hot deserts such as the Sahara.

Some sand dunes in the Sahara are more than 600 ft (180 m) tall.